

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20		VA5	MODULAR DBB	K15AS22	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV	0	0		USE ONLY FOR TIE IN
30			MODULAR DBB	K15AS26	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV	0	0	817	ALL BOLTS ARE METRIC DIA.
40		VAV	MODULAR DBB	K09AS26	3	3	MODULAR VALVE DB&B BALL TYPE 900# RF SS A182 F316 316SS TRIM METAL LO 718 BOLT FULL SIZE TYPE 2BV	0	0	B1	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
50			MODULAR DBB	M15AS22	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
60			MODULAR DBB	M15AS26	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR	0	0	B6	TO BE USED WITH RAISED FACE FLANGES ONLY.
70		VAW	MODULAR DBB	M15ES22	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	BA1	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. FOR FGP ONLY.
80		VAW	MODULAR DBB	M09ES22	1	1	MODULAR VALVE DB&B BALL TYPE 900# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	BA5	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
90	U04	VAW	MODULAR DBB	M09ES22	1	1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	FA8	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
100							***** BALL VALVES *****			FF5	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
100		VB2								FS1	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
110			BALL	A15AS25	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO 718 BOLT	0	0	SB6	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE AND ORIFICE SETS
110			BALL	A15AS26	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION METAL RB LO 718 BOLT	0	0	SC1	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.
120			BALL	A09AS25	3	3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO 718 BOLT	0	0	UD6	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST
130			BALL	A09AS26	3	3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL RB LO 718 BOLT	0	0	UD7	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
140			BALL	A09AS27	4	10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
150	U04		BALL	A09AS27	4	10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
160			BALL	A09AS28	4	10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT	0	0	VB2	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
170	U04		BALL	A09AS28	4	10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT	0	0	VC1	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
180			BALL	A09CS27	12	30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT	0	0		SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
190	U04	817	BALL	A09CS27	12	30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT	0	0		
200	U04	817	BALL	A09CS28	12	30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT	0	0		
210			BALL	B15AS21	0.75	1.5	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT	0	0		
220	U04	B6	BALL	B15AS22	0.75	1.5	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT	0	0		
230			BALL	B09AS25	3	3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB LO 718 BOLT	0	0		
240			BALL	B09AS26	3	3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL FB LO 718 BOLT	0	0		
250	U04	VA5	BALL	B09AS27	3	10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT	0	0		
260	U04	VA5	BALL	B09AS28	3	10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT	0	0		
270	U04	817	BALL	B09CS27	12	30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT	0	0		
280	U04	817	BALL	B09CS28	12	30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT	0	0		
290	U04	VA5	BALL	B15AS28	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT	0	0		
300	U04		BALL	A15AS28	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT	0	0		
310	U04	VA5	BALL	B15AS27	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT	0	0		
320	U04		BALL	A15AS27	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT	0	0		
330							***** GLOBE VALVES *****				
340			GLOBE	L15AS55	0.75	1.5	GLOBE 1500# RF A182 F316 SS TRIM 16 OS&Y HW 718 BOLT	0	0		
350			GLOBE	L15AS56	2	2	GLOBE 1500# RF SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT	0	0		
360			GLOBE	L09AS56	3	3	GLOBE 900# RF SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT 718 BOLT	0	0		
370			GLOBE	L09AS57	4	8	GLOBE 900# RF SS A351 CF8M TRIM 16 OS&Y GO 718 BOLT	0	0		
380	U04	817	GLOBE	L15RS55	1	1	GLOBE 1500# RTJ A182 F316 SS TRIM 16 OS&Y HW 718 BOLT	0	0		
390	U04	817	GLOBE	L15RS56	2	2	GLOBE 1500# RTJ SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT	0	0		
400	U04	817	GLOBE	L09RS56	3	3	GLOBE 900# RTJ SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT 718 BOLT	0	0		
410							***** CHECK VALVES *****				
420											
420		VC1	CHECK	C15AS54	0.75	1.5	CHECK PISTON 1500# RF SS A182 F316 TRIM 16 718 BOLT	0	0		
430	U04		CHECK	C15AS45	2	2	CHECK DUAL WAFER LUG 1500# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0		
440	U04		CHECK	C09AS45	3	8	CHECK DUAL WAFER LUG 900# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0		
450	U04		CHECK	C09AS52	10	10	CHECK DUAL DBL FLG 900# RF SS A351 CF8M TRIM 16	0	0		
460	U04		CHECK	C09CS52	12	30	CHECK DUAL DBL FLG 900# CF SS A351 CF8M TRIM 16	0	0		
470							***** PIPE *****				
480											
480	U04		PIPE	5365140	0.5	1.5	PIPE SCH 80S SMLS SS A312 TP316/316L PE	0	0		
490			PIPE	5365138	2	3	PIPE SCH 80S SMLS SS A312 TP316/316L BE	0	0		
500			PIPE	5365128	4	24	PIPE SCH 120 SMLS SS A312 TP316/316L BE	0	0		
510			PIPE	5365217	26	26	PIPE WT 42.0MM EFW SS A358 316/316L BE	0	0		
520	U04		PIPE	5365142	26	26	PIPE WT 48.0MM EFW SS A358 TP316/316L BE	0	0		
530			PIPE	5365218	28	28	PIPE WT 44.0MM EFW SS A358 316/316L BE	0	0		
540			PIPE	5365216	30	30	PIPE WT 47.0MM EFW SS A358 316/316L BE	0	0		
550			PIPE	5365219	36	36	PIPE WT 56.0MM EFW SS A358 316/316L BE	0	0		
560		814	PIPE	5365207	0.75	0.75	PIPE SCH 160 SMLS SS A312 TP316/316L PE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	NACE STD. MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -140 / 350 °C	12/16/2024	BY OZAC	U04	Page 1 of 5	900P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
570							***** FITTINGS *****							
570														
580														
580	U04		ELBOW	5449033	0.5	1.5	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S	0	0					
590			ELBOW	5449033	2	3	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S	0	0					
600			ELBOW	5449570	4	24	ELL 90 DEG LR SCH 120 SS A403 WP316/316L-S	0	0					
610			ELBOW	5451124	26	26	ELL 90 DEG LR WT 42.0MM SS A403 WP316/316L-W	0	0					
620			ELBOW	5451125	28	28	ELL 90 LR WT 44.0MM 316/316L SS A403 WP316/316L-W	0	0					
630			ELBOW	5451136	30	30	ELL 90 LR WT 47.0MM SS A403 WP316/316L-W	0	0					
640			ELBOW	5451127	36	36	ELL 90 LR WT 56.0MM SS A403 WP316/316L-W	0	0					
650	U04		ELBOW	5480138	0.75	1.5	ELL 90 DEG 3000# SW SS A182 F316/316L	0	0					
660	U04		ELBOW	5449035	4	24	ELL 90 DEG LR SCH 120 SS A403 WP316/316L-W	0	0					
670	U04		ELBOW	5449092	26	26	ELL 90 DEG LR WT 48.0MM SS A403 WP316/316L-W	0	0					
680	U04		ELBOW	5449146	0.5	3	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S	0	0					
690			ELBOW	5449576	4	24	ELL 45 DEG LR SCH 120 SS A403 WP316/316L-S	0	0					
700			ELBOW	5449084	26	26	ELL 45 DEG LR WT 42.0MM SS A403 WP316/316L-W	0	0					
710			ELBOW	5449085	28	28	ELL 45 DEG LR WT 44.0MM SS A403 WP316/316L-W	0	0					
720			ELBOW	5451126	30	30	ELL 45 LR WT 47.0MM SS A403 WP316/316L-W	0	0					
730	U04		ELBOW	5480139	0.75	1.5	ELL 45 DEG 3000# SW SS A182 F316/316L	0	0					
740	U04		ELBOW	5449165	4	24	ELL 45 DEG LR SCH 120 SS A403 WP316/316L-W	0	0					
750	U04		ELBOW	5449076	26	26	ELL 45 DEG LR WT 48.0MM SS A403 WP316/316L-W	0	0					
760	U04		ELBOW	5449086	36	36	ELL 45 LR WT 56.0MM SS A403 WP316/316L-W	0	0					
770														
770			TEE	5449156	0.75	3	TEE SCH 80S SS A403 WP316/316L-S	0	0					
780			TEE	5449579	4	24	TEE SCH 120 SS A403 WP316/316L-S	0	0					
790			TEE	5449087	26	26	TEE 42.0MM SS A403 WP316/316L-W	0	0					
800			TEE	5449088	28	28	TEE 44.0MM SS A403 WP316/316L-W	0	0					
810			TEE	5451116	30	30	TEE 47.0MM SS A403 WP316/316L-W	0	0					
820			TEE	5449089	36	36	TEE 56.0MM SS A403 WP316/316L-W	0	0					
830			TEE RED	5448905	0.75	2	TEE RED SCH 80S SS A403 WP316/316L-S	1	3					
840			TEE RED	5449577	2	3	TEE RED SCH 120/80S SS A403 WP316/316L-S	4	6					
850			TEE RED	5449578	4	20	TEE RED SCH 120 SS A403 WP316/316L-S	6	24					
860			TEE RED	5449094	12	24	TEE RED WT 42.0MM X SCH 120 SS A403 WP316/316L-W	26	26					
870			TEE RED	5449054	14	24	TEE RED WT 47.0MM X SCH 120 SS A403 WP316/316L-W	30	30					
880														
880			CAP	5449049	0.75	3	CAP SCH 80S SS A403 WP316/316L-S	0	0					
890	U04		CAP	5480141	0.75	1.5	CAP 3000# SW SS A182 F316/316L	0	0					
900			CAP	5449052	4	24	CAP SCH 120 SS A403 WP316/316L-S	0	0					
910			CAP	5449090	26	26	CAP WT 42.0MM SS A403 WP316/316L	0	0					
920	U04		CAP	5449079	26	26	CAP WT 48.0MM SS A403 WP316/316L	0	0					
930			CAP	5449091	28	28	CAP WT 44.0MM SS A403 WP316/316L	0	0					
940			CAP	5449097	30	30	CAP WT 47.0MM SS A403 WP316/316L	0	0					
950			CAP	5449093	36	36	CAP WT 56.0MM SS A403 WP316/316L	0	0					
960														
960	U04		RED CONC	5449158	0.5	2	REDUCER CONC SCH 80S SS A403 WP316/316L-S	0.75	3					
970	U04		RED CONC	5448916	1.5	3	REDUCER CONC SCH 120/80S SS A403 WP316/316L-S	4	6					
980	U04		RED CONC	5448915	4	20	REDUCER CONC SCH 120 SS A403 WP316/316L-S	6	24					
990			RED CONC	5449095	18	24	REDUCER CONC 42.0MM WT/SCH 120 SS A403 WP316/316L-W	26	26					
1000			RED CONC	5450766	24	24	REDUCER CONC 47.0MM / SCH 120 SS A403 WP316/316L-W	30	30					
1010			RED CONC	5450753	24	24	REDUCER CONC 56.0MM / SCH 120 SS A403 WP316/316L-W	36	36					
1020			RED CONC	5450757	30	30	REDUCER CONC 56.0MM/ 47.0MM SS A403 WP316/316L-W	36	36					
1030	U04		RED CONC	5449080	18	24	REDUCER CONC 48.0MM WT/SCH 120 SS A403 WP316/316L-W	26	26					
1040	U04		RED CONC	5448911	4	20	REDUCER CONC SCH 120 SS A403 WP316/316L-W	6	24					
1050	U04		RED ECC	5449159	0.5	2	REDUCER ECC SCH 80S SS A403 WP316/316L-S	0.75	3					
1060			RED ECC	5449179	1.5	3	REDUCER ECC SCH 120/80S SS A403 WP316/316L-S	4	6					
1070			RED ECC	5449178	4	20	REDUCER ECC SCH 120 SS A403 WP316/316L-S	6	24					
1080			RED ECC	5449096	18	24	REDUCER ECC 42.0MM WT/SCH 120 SS A403 WP316/316L-W	26	26					
1090			RED ECC	5450716	24	24	REDUCER ECC 47.0MM / SCH 120 SS A403 WP316/316L-W	30	30					
1100			RED ECC	5450755	24	24	REDUCER ECC 56.0MM / SCH 120 SS A403 WP316/316L-W	36	36					
1110			RED ECC	5450734	30	30	REDUCER ECC 56.0MM / 47.0MM SS A403 WP316/316L-W	36	36					
1120	U04		RED ECC	5449081	18	24	REDUCER ECC 48.0MM WT/SCH 120 SS A403 WP316/316L-W	26	26					
1130	U04		RED ECC	5449163	4	20	REDUCER ECC SCH 120 SS A403 WP316/316L-W	6	24					
1140	U04		RED ECC	5449162	2	3	REDUCER ECC SCH 120/80 SS A403 WP316/316L-W	4	6					
1150	U04		RED ECC	5449161	2	2	REDUCER ECC SCH 80 SS A403 WP316/316L-S	3	3					
1160														
1160			OLETS	5504176	0.75	1.5	NIPOFLANGE 1500# RF SCH 80S SS A182 F316/316L	2	3					
1170			OLETS	5504176	2	2	NIPOFLANGE 1500# RF SCH 80S SS A182 F316/316L	3	3					
1180			OLETS	5504177	0.75	2	NIPOFLANGE 1500# RF SCH 120/80S SS A182 F316/316L	4	24					
1190			OLETS	5504178	0.75	2	NIPOFLANGE 1500# RF 42.0MM WT/SCH 80S SS A182 F316/316L	26	26					
1200			OLETS	5504179	0.75	2	NIPOFLANGE 1500# RF 47.0MM WT/SCH 80S SS A182 F316/316L	30	30					
1210			OLETS	5504189	0.75	2	NIPOFLANGE 1500# RF 56.0MM WT/SCH 80S SS A182 F316/316L	36	36					
1220	UJ2		OLETS	5504180	1.5	1.5	NIPOFLANGE 1500# RF SCH 120/160 SS A182 F316/316L	4	24					
1230	UJ2		OLETS	5504181	1.5	1.5	NIPOFLANGE 1500# RF 42.0MM WT/SCH 160 SS A182 F316/316L	26	26					

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	NACE STD. MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -140 / 350 °C	12/16/2024	BY OZAC	U04	Page 2 of 5	900P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1240		UJ2	OLETS	5504182	1.5	1.5	NIPOFLANGE 1500# RF 47.0MM WT/SCH 160 SS A182 F316/316L	30	30	
1250		UJ2	OLETS	5504188	1.5	1.5	NIPOFLANGE 1500# RF 56.0MM WT/SCH 160 SS A182 F316/316L	36	36	
1260	U04	UJ2	OLETS	5504444	1.5	1.5	NIPOFLANGE 1500# RTJ 42.0MM WT/SCH 160 SS A182 F316/316L	26	26	
1270	U04		OLETS	5504517	2	2	NIPOFLANGE 1500# RTJ 42.0MM WT/SCH 80 SS A182 F316/316L	26	26	
1280	U04		OLETS	5504443	0.75	1.5	NIPOFLANGE 1500# RTJ 42.0MM WT/SCH 80S SS A182 F316/316L	26	26	
1290	U04	UJ2	OLETS	5504445	1.5	1.5	NIPOFLANGE 1500# RTJ 47.0MM WT/SCH 160 SS A182 F316/316L	30	30	
1300	U04		OLETS	5504514	2	2	NIPOFLANGE 1500# RTJ 47.0MM WT/SCH 80 SS A182 F316/316L	30	30	
1310	U04		OLETS	5504448	0.75	1.5	NIPOFLANGE 1500# RTJ 47.0MM WT/SCH 80S SS A182 F316/316L	30	30	
1320	U04	UJ2	OLETS	5504442	1.5	1.5	NIPOFLANGE 1500# RTJ 48.0MM WT/SCH 160 SS A182 F316/316L	26	26	
1330	U04		OLETS	5504512	2	2	NIPOFLANGE 1500# RTJ 48.0MM WT/SCH 80 SS A182 F316/316L	26	26	
1340	U04		OLETS	5504441	0.75	1.5	NIPOFLANGE 1500# RTJ 48.0MM WT/SCH 80S SS A182 F316/316L	26	26	
1350	U04	UJ2	OLETS	5504440	1.5	1.5	NIPOFLANGE 1500# RTJ SCH 120/160 SS A182 F316/316L	4	24	
1360	U04		OLETS	5504511	2	2	NIPOFLANGE 1500# RTJ SCH 120/80 SS A182 F316/316L	4	24	
1370	U04		OLETS	5504040	0.75	1.5	NIPOFLANGE 1500# RTJ SCH 120/80S SS A182 F316/316L	4	24	
1380	U04		OLETS	5504510	2	2	NIPOFLANGE 1500# RTJ SCH 80 SS A182 F316/316L	3	3	
1390	U04		OLETS	5504038	0.75	1.5	NIPOFLANGE 1500# RTJ SCH 80/80S SS A182 F316/316L	2	3	
1400	U04	FF5	OLETS	5504058	0.75	0.75	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 80S SS A182 F316/316L	0	0	
1410										
1410			OLETS	5509637	0.75	1	WELDOLET SCH 80S SS A182 F316/316L	2	3	
1420			OLETS	5510374	0.75	3	WELDOLET SCH 120 /80S SS A182 316/316L	4	24	
1430			OLETS	5509649	2	3	WELDOLET 42.0MM WT/SCH 80S SS A182 F316/316L	26	26	
1440			OLETS	5509647	2	3	WELDOLET 47.0MM WT/SCH 80S SS A182 F316/316L	30	30	
1450			OLETS	5509640	4	8	WELDOLET SCH 120 SS A182 F316/316L	12	24	
1460			OLETS	5509646	4	10	WELDOLET 42.0MM WT/SCH 120 SS A182 F316/316L	26	26	
1470			OLETS	5509648	4	12	WELDOLET 47.0MM WT/SCH 120 SS A182 F316/316L	30	30	
1480										
1480		UJ2	ELBOFLG	5516085	1.5	1.5	ELBOFLANGE 1500# RF SCH 120/160 SS A182 F316/316L	4	24	
1490		UJ2	ELBOFLG	5516086	1.5	1.5	ELBOFLANGE 1500# RF 42.0MM WT/SCH 160 SS A182 F316/316L	26	26	
1500		UJ2	ELBOFLG	5516087	1.5	1.5	ELBOFLANGE 1500# RF 47.0MM WT/SCH 160 SS A182 F316/316L	30	30	
1510		UJ2	ELBOFLG	5516088	1.5	1.5	ELBOFLANGE 1500# RF 56.0MM WT/SCH 160 SS A182 F316/316L	36	36	
1520	U04	UJ2, 817	ELBOFLG	5516443	1.5	1.5	ELBOFLANGE 1500# RTJ 42.0MM WT/SCH 160 SS A182 F316/316L	26	26	
1530	U04	UJ2, 817	ELBOFLG	5516445	1.5	1.5	ELBOFLANGE 1500# RTJ 47.0MM WT/SCH 160 SS A182 F316/316L	30	30	
1540	U04	UJ2, 817	ELBOFLG	5516442	1.5	1.5	ELBOFLANGE 1500# RTJ 48.0MM WT/SCH 160 SS A182 F316/316L	26	26	
1550	U04	UJ2, 817	ELBOFLG	5516440	1.5	1.5	ELBOFLANGE 1500# RTJ SCH 120/160 SS A182 F316/316L	4	24	
1560										
1560							*****COUPLINGS,PLUG,UNION*****			
1570										
1570			COUPLING	5480142	0.5	1.5	COUPLING 3000# SW SS A182 F316/316L	0	0	
1580			TEE RED	5480134	0.75	1	COUPLING RED 3000# SW SS A182 F316/316L	1	1.5	
1590										
1590			UNION	5480146	0.75	1.5	UNION 3000# SW SS A182 F316/316L	0	0	
1600										
1600		UD6	CAP	5498274	0.75	0.75	CAP 3000# SCRD A182 F316/316L	0	0	
1610										
1610		UD7	FITTING SCRD	5498383	0.75	0.75	PLUG ROUND HEAD SCRD SS A182 F316/316L	0	0	
1620										
1620							*****OLETS*****			
1630										
1630			OLETS	5537071	0.75	1.5	SOCKOLET 3000# SS A182 F316/316L	2	26	
1640			OLETS	5533235	0.75	1.5	SOCKOLET FLAT BOTTOM 3000# SS A182 F316/316L	0	0	
1650										
1650		UD7	OLETS	5515771	0.75	0.75	THREDOLET 3000# SS A182 F316/316L	2	30	
1660										
1660		816					***** FLANGES *****			
1670										
1670		FA8	TEE RED	5600664	0.5	0.5	FLG SCRD (FNPT) REDUCING 1500# RF SS A182 F316/316L	0.75	1.5	
1680	U04	SC1, 817	FLANGE	5600662	0.75	0.75	FLG SCRD (FNPT) 1500# RTJ SS A182 F316/316L	0	0	
1690	U04	FA8, 817	FLANGE	5600661	0.5	0.5	FLG SCRD (FNPT) REDUCING 1500# RTJ SS A182 F316/316L	0.75	0.75	
1700	U04	FA8, 817	FLANGE	5600661	0.5	0.5	FLG SCRD (FNPT) REDUCING 1500# RTJ SS A182 F316/316L	1.5	1.5	
1710										
1710			FLANGE	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
1720			FLANGE	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
1730										
1730	U04		FLANGE	5583979	0.5	2	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE	0	0	
1740			FLANGE	5583006	3	3	FLG WN 900# RF SS A182 F316/316L SCH 80S BORE	0	0	
1750			FLANGE	5583813	4	10	FLG WN 900# RF SS A182 F316/316L SCH 120 BORE	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	NACE STD. MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -140 / 350 °C	12/16/2024	BY OZAC	U04	Page 3 of 5	900P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1760			FLANGE	5583813	12	24	FLG WN 900# RF SS A182 F316/316L SCH 120 BORE	0	0	
1770			FLANGE	5583018	26	26	FLG WN 900# RF SS A182 F316/316L 42.0MM WT B16.47 SERIES A	0	0	
1780			FLANGE	5583019	30	30	FLG WN 900# RF SS A182 F316/316L 47.0MM WT B16.47 SERIES A	0	0	
1790			FLANGE	5583020	36	36	FLG WN 900# RF SS A182 F316/316L 56.0MM WT B16.47 SERIES A	0	0	
1800	U04	817	FLANGE	5608765	0.75	1.5	FLG SW 1500# RTJ SS A182 F316/316L SCH 80S BORE	0	0	
1810	U04	817	FLANGE	5584065	6	6	FLG WN 1500# RTJ SS A182 F316/316L SCH 120 BORE	0	0	
1820	U04	817	FLANGE	5583000	2	2	FLG WN 1500# RTJ SS A182 F316/316L SCH 80 BORE	0	0	
1830	U04	817	FLANGE	5584061	0.75	1	FLG WN 1500# RTJ SS A182 F316/316L SCH 80S BORE	0	0	
1840	U04	FS1, 817	FLANGE	5584167	0.5	0.5	FLG WN 2500# RTJ SS A182 F316/316L SCH 80S	0	0	
1850	U04	FS1, 817	FLANGE	5584167	1.5	1.5	FLG WN 2500# RTJ SS A182 F316/316L SCH 80S	0	0	
1860	U04	817	FLANGE	5583012	26	26	FLG WN 900# RTJ SS A182 F316/316L 42.0MM WT B16.47 SERIES A	0	0	
1870	U04	817	FLANGE	5583631	30	30	FLG WN 900# RTJ SS A182 F316/316L 47.0MM WT B16.47 SERIES A	0	0	
1880	U04	817	FLANGE	5583011	26	26	FLG WN 900# RTJ SS A182 F316/316L 48.0MM WT B16.47 SERIES A	0	0	
1890	U04	817	FLANGE	5583002	4	4	FLG WN 900# RTJ SS A182 F316/316L SCH 120 BORE	0	0	
1900	U04	817	FLANGE	5583002	6	24	FLG WN 900# RTJ SS A182 F316/316L SCH 120 BORE	0	0	
1910	U04	817	FLANGE	5583001	3	3	FLG WN 900# RTJ SS A182 F316/316L SCH 80 BORE	0	0	
1920	U04	817	FLANGE	5619824	12	24	FLG COMPACT 900# RJ SS A182 F316/316L SCH 120 BORE	0	0	
1930	U04	817	FLANGE	5619825	26	26	FLG COMPACT 900# RJ SS A182 F316/316L WT 42.0MM	0	0	
1940	U04	817	FLANGE	5619826	28	28	FLG COMPACT 900# RJ SS A182 F316/316L WT 44.0MM	0	0	
1950	U04	817	FLANGE	5619827	30	30	FLG COMPACT 900# RJ SS A182 F316/316L WT 47.0MM	0	0	
1960	U04	817	FLANGE	5619828	36	36	FLG COMPACT 900# RJ SS A182 F316/316L WT 56.0MM	0	0	
1970			BLIND	5616564	0.5	2	FLG BLIND 1500# RF SS A182 F316/316L	0	0	
1980			BLIND	5616556	3	10	FLG BLIND 900# RF SS A182 F316/316L	0	0	
1990			BLIND	5616556	12	24	FLG BLIND 900# RF SS A182 F316/316L	0	0	
2000			BLIND	5616533	26	30	FLG BLIND 900# RF SS A182 F316/316L B16.47 SERIES A	0	0	
2010	U04	817	BLIND	5616567	0.75	2	FLG BLIND 1500# RTJ SS A182 F316/316L	0	0	
2020	U04	817	BLIND	5616558	3	24	FLG BLIND 900# RTJ SS A182 F316/316L	0	0	
2030	U04	817	BLIND	5616532	26	30	FLG BLIND 900# RTJ SS A182 F316/316L B16.47 SERIES A	0	0	
2040	U04	817	BLIND	5619752	12	36	FLG COMPACT BLIND 900# RJ SS A182 F316/316L LINE CLASS 900P21	0	0	
2050							***** ORIFICE FLANGES *****			
2060										
2060			ORIFICE FLG	5641519	2	2	WN ORIF SET 1500# RF SS A182 F316/316L SCH 80S BORE	0	0	
2070			ORIFICE FLG	5641287	3	3	WN ORIF SET 900# RF SS A182 F316/316L SCH 80S BORE	0	0	
2080			ORIFICE FLG	5641293	4	24	WN ORIF SET 900# RF SS A182 F316/316L SCH 120 BORE	0	0	
2090			ORIFICE FLG	5640609	26	26	WN ORIF SET 900# RF A182 F316/316L 42.0MM WT B16.47 SERIES A	0	0	
2100			ORIFICE FLG	5640614	30	30	WN ORIF SET 900# RF A182 F316/316L 47.0MM WT B16.47 SERIES A	0	0	
2110							***** PIPE NIPPLES *****			
2110										
2120										
2120	U04		NIPPLE	5542154	0.75	1.5	NIPPLE SCH 160 SS A312 TP316/316L TBE 3" LG	0	0	
2130	U04		NIPPLE	5542155	0.75	1.5	NIPPLE SCH 160 SS A312 TP316/316L TBE 7" LG	0	0	
2140	U04		NIPPLE	5542164	0.75	1.5	NIPPLE SCH 160/80S A312 TP316/316L TOE/POE 3 IN LONG	0	0	
2150	U04		NIPPLE	5542165	0.75	1.5	NIPPLE SCH 160/80S A312 TP316/316L TOE/POE 7 IN LONG	0	0	
2160							***** GASKETS *****			
2160										
2170										
2170			GASKET	5672251	0.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2180			GASKET	5673117	3	24	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2190			GASKET	5674033	26	36	GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
2200	U04	817	GASKET	5673450	12	36	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900P21	0	0	
2210	U04	817	GASKET	5674479	0.75	2	GASKET 1500# RTJ SS OCTAGONAL RING	0	0	
2220	U04	817	GASKET	5674479	6	6	GASKET 1500# RTJ SS OCTAGONAL RING	0	0	
2230	U04	FS1, 817	GASKET	5674480	0.5	0.5	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
2240	U04	FS1, 817	GASKET	5674480	1.5	1.5	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
2250	U04	817	GASKET	5674481	3	24	GASKET 900# RTJ SS OCTAGONAL RING	0	0	
2260	U04	817	GASKET	5674485	26	30	GASKET 900# RTJ SS OCTAGONAL RING ASME B16.47 SERIES A	0	0	
2270							***** BOLTING *****			
2270		B1								
2280										
2280				5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2290		BA1	BOLTING	5675677	0.5	10	STUD-BOLT A320 GR B8M CLASS 2/A194 GR 8MA NUTS	0	0	
2300	U04	BA1	BOLTING	5675677	12	99	STUD-BOLT A320 GR B8M CLASS 2/A194 GR 8MA NUTS	0	0	
2310	U04		BOLTING	5675776	12	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
2320			BOLTING	5675776	0.5	10	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
2330	U04	BA5	BOLTING	5675778	12	99	STUD-BOLT C/W 2 COMPACT NUTS API-6A718	0	0	
2340										
2340		SB6	BOLTING	5675676	10	30	JACK SCREW HEX HEAD A320 B8M - CLASS 2	0	0	
2350										

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SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2350							***** BLINDS AND SPACERS *****			
2360										
2360			SPEC BLIND	5749757	1	2	SPECTACLE BLIND 1500# RF SS A240 GR 316	0	0	
2370			SPEC BLIND	5749755	3	6	SPECTACLE BLIND 900# RF SS A240 GR 316	0	0	
2380										
2380			PADDLE	5750048	8	10	PADDLE BLANK 900# RF SS A240 GR 316	0	0	
2390			PADDLE	5750048	12	24	PADDLE BLANK 900# RF SS A240 GR 316	0	0	
2400			PADDLE	5750282	8	10	PADDLE SPACER 900# RF SS A240 GR 316	0	0	
2410			PADDLE	5750282	12	24	PADDLE SPACER 900# RF SS A240 GR 316	0	0	
2420			PADDLE	5750532	26	36	PADDLE BLANK 900# RF SS A240 GR 316 B16.47 SERIES A	0	0	
2430			PADDLE	5750632	26	36	PADDLE SPACER 900# RF SS A240 GR 316 B16.47 SERIES A	0	0	
2440	U04	817	PADDLE	5750274	12	36	PADDLE BLANK COMPACT 900# RJ SS A240 GR 316	0	0	
2450	U04	817	PADDLE	5750275	12	36	PADDLE SPACER COMPACT 900# RJ SS A240 GR 316	0	0	
2460										

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